

Topic 2.14 - Logarithmic Function Context and Data Modeling

Practice Worksheet

Strengthened ECHS edition - original reasoning, modeling, and AP-style tasks

Name: _____ Class: _____ Date: _____

Directions

Show a mathematical argument, not only a final answer. Use exact values unless an approximation is requested. Calculator use is identified on each item. For contextual models, state units, domain restrictions, and assumptions when relevant.

Learning focus

- Construct logarithmic models from proportional input change, two points, transformations, or regression output.
- Use logarithmic models to predict outputs and solve for inputs.
- Interpret domains, units, scale meaning, diminishing change, and model limitations.

Section A: Foundation and representation

Question 1

Core skill**No calculator**

A response is 18 at $x = 1$ and increases by 7 whenever x is multiplied by 4. Write a model.

Question 2

Core skill**No calculator**

Construct $f(x) = a \log_3 x + k$ through $(1, -2)$ and $(27, 10)$.

Question 3

Core skill

No calculator

A model has vertical asymptote $x = 5$, anchor point $(6, 11)$, and output increases by 8 whenever $x - 5$ is multiplied by 2. Write the model.

Question 4

Core skill

No calculator

For $S(x) = 24 + 6 \ln x$, find $S(1)$, $S(e^2)$, and the change from x to e^3x .

Question 5

Core skill

Calculator permitted

Solve $40 + 5 \log_{10} x = 57$ for x .

Question 6

Interpretation

Calculator permitted

A fitted model is $y = 72 + 9.5 \ln x$. Interpret the coefficient 9.5 in terms of multiplying the input by e and by 10.

Section B: Application, comparison, and error analysis

Question 7

Context

No calculator

A learning model is $L(h) = 45 + 12 \log_2(h + 1)$, where $h \geq 0$ is practice hours. Find the predicted gain when practice increases from 3 to 7 hours.

Question 8

Scale reasoning

No calculator

A model $D(x) = 20 \log(x/x_0)$ uses a positive reference x_0 . Show that multiplying x by 100 increases D by 40.

Question 9

Error analysis

No calculator

A student fits $y = a + b \ln x$ to data containing $x = 0$. Explain the problem and suggest one possible remedy.

Question 10

Comparison

Calculator permitted

Compare $f(x) = 5 \ln x$ and $g(x) = 2 \log_2 x$ by rewriting both with natural logarithms. Which grows faster as a function of $\ln x$?

Question 11

Limitation

No calculator

A logarithmic model predicts an output of -40 near its positive domain boundary, but the measured quantity cannot be negative. What should be concluded?

Question 12

Construction

No calculator

Construct a logarithmic model that passes through $(3, 4)$ and $(11, 16)$ and has vertical asymptote $x = 2$, using base 3.

Section C: AP-style multiple choice**MCQ 1**

No calculator

A model increases by 5 whenever the input is multiplied by 2. Which form is consistent?

- A. $5x + 2$
- B. $5 \log_2 x + k$
- C. $2^x + 5$
- D. $k(5)^x$

MCQ 2

No calculator

For $f(x) = a \ln x + k$, multiplying x by e^3 changes f by:

- A. 3
- B. $a + 3$
- C. $3a$
- D. e^{3a}

MCQ 3

No calculator

The domain of $7 + 4\log(x - 6)$ is:

- A. $x < 6$
- B. $x > 6$
- C. $x \neq 6$
- D. All real x

MCQ 4

No calculator

Which statement is a valid model limitation?

- A. Logarithmic functions can never model data.
- B. A high fit statistic proves causation.
- C. The formula may be undefined or unrealistic near the domain boundary.
- D. Logarithmic growth becomes negative for all large inputs.

Section D: Free-response reasoning**FRQ 1 - Construct a logarithmic model**

No calculator

6 points

A response model has vertical asymptote $x = 2$. It passes through $(3, 10)$ and $(11, 22)$, and uses base 3. (a) Construct the model. (b) State its domain and interpret the anchor point. (c) Find the input that gives output 16. (d) Describe the output change when $x - 2$ is multiplied by 27.

FRQ 2 - Evaluate a fitted model

Calculator permitted

6 points

A fitted model is $S(x) = 38 + 7.2 \ln x$ for $1 \leq x \leq 50$. (a) Interpret $S(1)$ and the coefficient 7.2. (b) Predict $S(10)$. (c) Find the input that gives $S = 60$. (d) Explain why predictions at $x = 0.1$ and $x = 5000$ require caution.
